

Assignment : Multimodal Learning with Limited Labels and Missing Modalities

1 Objective

The objective of this assignment is to explore multimodal learning approaches using a dataset containing images, text, and their associated labels. You will train models to investigate the impact of 1) limited data annotations during training and 2) missing modalities during inference on classification performance. Results must include Top 1 and Top 5 accuracy metrics. You will develop algorithms to improve the model performance under these two scenarios.

1.1 Dataset

You have been provided with two datasets: COCO Captions and the UPMC Food-101 dataset. You can use one or both of them for this assignment.

The COCO (Common Objects in Context) Captions [1] dataset is a large-scale dataset commonly used for object detection, segmentation, and captioning tasks in computer vision. It contains images of everyday scenes, each annotated with object bounding boxes, object categories, and descriptive captions. The captions provide detailed descriptions of the objects and their relationships within the images, making the COCO dataset valuable for training and evaluating models in various visual understanding tasks. For this assignment, we've curated a selection of images alongside their respective captions and a set of labels intended for classification. You can find more details in the dataset folder.

UPMC Food-101 [2] is a multimodal classification dataset consisting of text and image pairs, with 101 food categories as labels. This dataset includes textual descriptions of 101 distinct food recipes complemented with images obtained from Google Image Search. The dataset consists of 67,972 training and 22,716 testing image-text pairs. The classification task involves utilizing the training set to train a model capable of accurately predicting the food labels for each image-text pair in the test set.

2 Tasks

Download the dataset from Canvas and use an appropriate dataloader with necessary preprocessing steps such as resizing images and text tokenization.

2.1 Unimodal classification performance:

1. Use the pretrained CLIP image encoder and train a model using 100% and 10% of the provided labels in the image training set. Report the Top 1 and Top 5 classification accuracy on image test set. **(5 pts)**
2. Use the pretrained CLIP text encoder and train a model using 100% and 10% of the provided labels in the text training set. Report the Top 1 and Top 5 classification accuracy on text test set. **(5 pts)**

2.2 Multimodal classification performance:

3. With pretrained CLIP encoders and an appropriate data fusion method, develop a model for predicting class labels using both visual and textual modalities. Train two models utilizing 100% and 10% of the provided labels in the training set, respectively. Evaluate each model's performance by reporting the Top 1 and Top 5 classification accuracy on the test set.
(5 pts)
4. Compare and discuss the results from the three previous tasks, providing explanations for any differences observed in performance of different models. **(5 pts)**
5. For any of the above models — image only, text only, or multimodal — suggest and implement a solution to improve classification performance when only 10% of training labels are accessible, potentially leveraging one of the self-supervised approaches discussed in lectures. Compare the results and discuss any limitations you encountered. **(30 pts)**

2.3 Modality Ablation:

6. Take your best performing multimodal model trained with all available labels in training set. Test the multimodal model performance while removing one modality during testing (first images and then text). Report Top 1 and Top 5 classification performance on the test set and discuss the results comparing with the scenario where both modalities are available during inference. **(15 pts)**

2.4 Robust Multimodal Learning:

7. Develop a multimodal learning approach capable of handling missing modalities during testing. Present a diagram and discuss its components. **(10 pts)**
8. Implement the proposed approach, train a model on the multimodal dataset, and evaluate classification performance with different combinations of available modalities. Discuss the results, limitations and suggest ideas for further improvement or future work. **(25 pts)**

3 Deliverables

- Python scripts containing the implementation of all tasks with thorough commenting.
- A report file containing answers to the questions and discussing the methodology, experimental results, and conclusions.

4 Evaluation Criteria

- Quality of responses and details of the selected models/approaches.
- Comprehensive analysis of the results.
- Clarity and coherence of the report.
- Accuracy improvements and efficient implementation.

You can find a detailed CRA on Canvas within the assignment submission module.

5 Additional Notes

- Data required for this assessment is available on canvas alongside this document in Assessment Details.
- Students are encouraged to experiment with different architectures, hyperparameters, and fusion techniques. You can also use any code presented in lectures/practicals/examples.
- Students are to work individually and submit their answers including a single report file (word document) and Python code with adequate comments via Canvas.
- Responses should explain and justify the approach taken to address the question including why the approach was selected over other possible methods. This explanation should take into account the nature of the problem, any data constraints, and computational/runtime factors. When evaluating the results, make sure to highlight any differences in performance between methods and discuss the strengths and weaknesses of each approach. And of course, include results, relevant figures, and analysis to provide a comprehensive analysis. Keep in mind that simply providing Python Notebooks or similar materials won't be sufficient as a response on their own.
- Responses should highlight where the modelling approach has been successful, and to highlight the limitations apparent. When limitations are encountered, students are encouraged to investigate and identify what the underlying cause of any failure or limitations are.

References

- [1] X. Chen, H. Fang, T.-Y. Lin, R. Vedantam, S. Gupta, P. Dollár, C. L. Zitnick, Microsoft coco captions: Data collection and evaluation server, arXiv preprint arXiv:1504.00325 (2015).
- [2] X. Wang, D. Kumar, N. Thome, M. Cord, F. Precioso, Recipe recognition with large multi-modal food dataset, in: 2015 IEEE International Conference on Multimedia & Expo Workshops (ICMEW), IEEE, 2015, pp. 1–6.